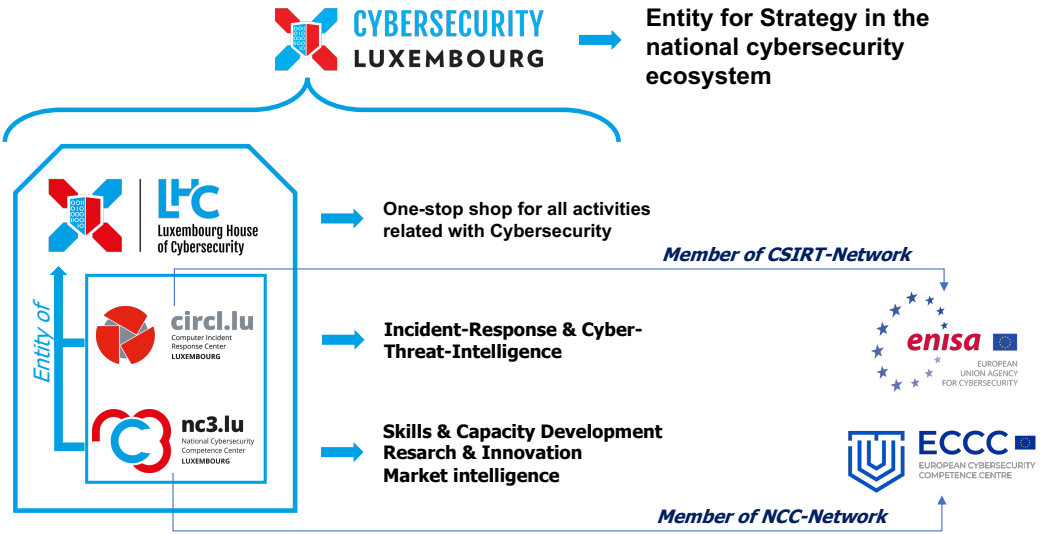
Range42 – Progress & Live Demo

Open Cyber Range Platform for Collaborative Security Training

NC3 / Range42 Team

FIC Lille 2026

Proxmox Ansible Orchestration Telemetry


CYBERSECURITY
LUXEMBOURG
Entity for Strategy in the national cybersecurity ecosystem
LHC
Luxembourg House of Cybersecurity
One-stop shop for all activities related with Cybersecurity
Member of CSIRT-Network
Entity of
circl.lu
Computer Incident Response Center
LUXEMBOURG
Incident-Response & Cyber-Threat-Intelligence
enisa
EUROPEAN UNION AGENCY FOR CYBERSECURITY
nc3.lu
National Cybersecurity Competence Center
LUXEMBOURG
Skills & Capacity Development
Resarch & Innovation
Market intelligence
ECCC
EUROPEAN CYBERSECURITY COMPETENCE CENTRE
Member of NCC-Network

Skills development

Capacity strengthening

Research & innovation

Ecosystem & industrialization

NCC coordination

ABOUT US

The mission of NC3 is to support the Luxembourg ecosystem in the development of skills and capacities in cybersecurity, thereby contributing to the development of an industrial base in cybersecurity and strengthening the strategic autonomy of the European Union.

nc3.lu
National Cybersecurity Competence Center
LUXEMBOURG

Public Money, Public Code. Let’s go all the way, shall we?

Team & Funding

- Core team of 3 – InfoSec & DevOps engineers
- Collaboration model with NC3 & ecosystem partners
- Public grant for cyber training infrastructure
- FOSS model: no licensing 💵, transparent development

Public Money

Public Code

publiccode.eu

Agenda

1. What Range42 is & why it matters
2. Progress since 2025: what's new
3. Architecture overview
4. Live demo: the Deployer UI in action
5. What's possible today vs. what's coming
6. Development roadmap: 6 strategic axes
7. How to get involved

What is Range42?

- **Open cyber range platform** for offensive, defensive, and hybrid training
- **Reproducible Infrastructure-as-Code**: Proxmox, Ansible, Docker
- **Flexible & extensible**: supports CVE labs, misconfigurations, forensics, malware analysis

> ℹ Built to simulate *real-world incidents* safely, with isolation, snapshots, and telemetry.

Progress Since 2025

Progress Since 2025

By the Numbers

- 520+ total commits across 14 repositories (up from 400+)
- Backend API: 156 commits – **major refactor** (86 routes → 10 modules)
- Deployer UI: 74 commits – from prototype to **functional tool**
- Proxmox controller: 112 commits – cloud-init, VM templates, storage

Key Milestones

- Docker deployment for the backend API
- Real-time VM status via WebSocket on the canvas
- Comprehensive test suite and Sphinx documentation
- CyCon 2026 paper submitted – academic publication

Current Achievements

What's Working Today

- ~100 CVEs & misconfigurations identified, ~20 deployable scenarios
- **Automated provisioning** on Proxmox with networking, VPN, firewalling
- **Visual lab designer** with drag-and-drop infrastructure composition
- **Integrated monitoring** via Wazuh for telemetry and alerting
- **14 repositories** managing automation, content, and tooling

> **Key Milestone**: Backend is production-ready. The visual designer is now functional and actively used.

Range42 vs. Other Cyber Ranges

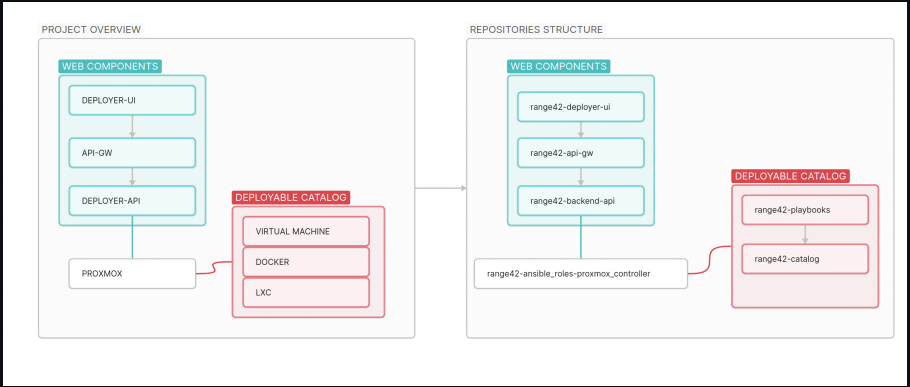
Feature	Range42	Commercial SaaS	Cloud Native	Traditional
Open Architecture	✓	×	×	×
IaC/GitOps	✓	~	✓	×
Private Deployment	✓	×	~	✓
Cost Control	✓	×	~	✓
Full Data Custody	✓	×	×	✓
API Orchestration	✓	✓	✓	×
Rapid Reset/Snapshots	✓	✓	~	~
Custom Scenarios	✓	×	~	✓
Visual Lab Designer	✓	✓	×	×

Range42's Edge: Full control, reproducibility, and cost-effectiveness without vendor lock-in.

Architecture at a Glance

- **Hypervisor layer**: Proxmox VMs/LXCs; snapshots; network segments
- **Automation layer**: Ansible roles orchestrate lifecycle, network, firewall, images
- **Control plane**: Backend API (FastAPI) with Docker deployment; WebSocket for live status
- **UX layer**: Deployer UI (visual designer), Exercise Management Platform (in development)
- **Observability**: Wazuh for logs/alerts; structured telemetry

Architecture: Logical Components → Repository Mapping



Left: Logical architecture flow. **Right**: Actual GitHub repository structure.

Live Demo: Deployer UI

Deployer UI: Visual Lab Designer

Design

- Drag-and-drop VMs, LXC, networks, routers, firewalls
- Configure CPU, memory, disk, ISO per node
- Proxmox template selection with auto-fill
- Connect nodes to define topology

Deploy

- Pre-deploy reconciliation and validation
- Ordered: networks → routers → VMs → config → start
- Real-time status via WebSocket
- Start, stop, restart, delete from the UI

Live Demo

Switch to live demo

Demo Flow

1. Create a new project on the canvas
2. Add VMs, networks, and a router
3. Configure VM templates and resources
4. Deploy to Proxmox
5. Watch real-time status updates
6. Control VMs from the UI (start/stop)

Backend API: Production-Ready

Major Refactoring

- 86 route files → **10 domain modules**
- 52 schema files → grouped modules
- Application factory pattern
- Structured logging throughout
- Centralized secret management

New Capabilities

- Docker deployment (compose)
- WebSocket API for live status
- Test suite: unit + integration
- Sphinx documentation

Today & Tomorrow

Today vs. Tomorrow

Operational Today

- Full Proxmox automation (VMs, LXC, network, firewall, VPN)
- Visual lab designer with drag-and-drop
- Real-time VM status via WebSocket
- Template provisioning + reconciliation
- Docker-based API deployment
- Wazuh monitoring integration

Coming Next

- Exercise Management Platform (scoring & reporting)
- LLM-assisted scenario generation
- Multi-subnet enterprise topologies
- Declarative Live State Engine
- ICS / IoT training scenarios
- Federated catalog exchange

Development Roadmap: 6 Strategic Axes

Axis 0 – Core Stabilisation & Hardening
Lock architecture, expand catalog to 50+ labs

Axis 1 – State Management (DLSE)
Declarative Live State Engine for reconciliation

Axis 2 – End-User Interfaces
Deployer UI, EMP with scoring dashboards

Axis 3 – Student-Facing Services
Portals, guided workflows, self-service access

Axis 4 – Industrial & IoT
ICS/SCADA and IoT training scenarios

Axis 5 – Open-Source Ecosystem
NGSOTI synergy, federated catalogs, community

Based on the Range42 Execution Framework v0.8 – 50 work blocks across 6 axes with quarterly sequencing.

Key Repositories

Organization Overview

14 Repositories (10 public, 4 private) – 520+ commits, zero high-severity findings

Repository	Commits	Language	Status
backend-api	156	Python / FastAPI	Active
proxmox_controller	112	Ansible / YAML	Active
catalog	97	Ansible / Docker	Stable
playbooks	81	Ansible / YAML	Active
deployer-ui	74	Vue 3 / TypeScript	Active

Lessons Learned

Lessons Learned 💡

Technical

- Ansible + Proxmox API = powerful IaC combo
- Snapshots are critical for training resets
- Early refactoring pays off (86 → 10 route files)
- WebSocket enables compelling real-time UX

Process

- Governance overhead is real but unlocks collaboration
- 6-axis framework aligns a small team
- Documentation lags code – a universal pattern
- Docker simplifies cross-environment deployment

Open Challenges & Opportunities ⚠

Content & Coverage

- Expand to 50+ deployable scenarios
- ICS/IoT and forensics content
- Standardized scenario format

Platform Maturity

- CI/CD across all repositories
- Exercise Management Platform
- Federated catalog exchange (NGSOTI)

👥 **Your Expertise Matters**: Every contribution – code, docs, testing, scenarios – moves the platform forward.

Get Involved

Contribute

- **Scenarios**: CVE research, automation, gamification
- **Infrastructure**: Ansible roles, backend API, CI/CD
- **UI/UX**: Lab designer, Exercise Management Platform

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Typeset with LaTeX

Contribute